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End-End communication at transport layer

Aim:

To implement echo client server using TCP/UDP sockets.

TCP

```
import socket
import threading

# .... Sower .....
def start_server():
    server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    server.bind(('127.0.0.1', 5000))
    server.listen(1)
    print("[SERVER] Listening on 127.0.0.1:5000")

    conn, addr = server.accept()
    print(f"[SERVER] connected by {addr}")

    while true:
        data = conn.recv(1024).decode()
        if not data or data.lower() == 'quit':
            print("[SERVER] Connection closed by client.")
            break
        print(f"[SERVER] Received: {data}")
        conn.send(f"Sower received: {data}".encode())

    conn.close()
    server.close()
```

```
#.... client .....
def start_client():
    client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    client.connect(('127.0.0.1', 5000))
    print("[CLIENT] connected to server")

while True:
    msg = input("Enter message (or 'quit' to exit): ")
    client.send(msg.encode())
    if msg.lower() == 'quit':
        break
    response = client.recv(1025)

client.close()

#..... MAIN ......
if __name__ == "__main__":
    #Run server in a background thread.
    server_thread = threading.Thread(target=start, daemon=True)
    server_thread.start()
    #Run client in main thread
    start_client()
```

output:-

[SERVER] Listening on 127.0.0.1:5000 [CLIENT] connected to server
[SERVER] connected ('127.0.0.1', 53458) Enter message (or 'quit' to exit):

hello
[Server] Received : hello
[Server Response] Server received : hello
Enter message (or 'quit' to exit) : quit
[Server] connection closed by client

UDP:

```
import Socket
import threading

def udp_Server():
    S=Socket.Socket(Socket.AF_INET,)
    S.bind('127.0.0.1', 12000))
print("[SERVER] Running on 127.0.0.1: 12000")
while true:
    data, addr=S.recvform(1024)
    msg=data.decode()
    if msg.lower()=='quit':
        printf(f"[Server] client {addr} ended th
connection.")
        break.
    print(f"[Server] Received from {addr}: {msg}")
S.close()
print("[SERVER] Stopped")

#....Client....
def_up_client():
    C=Socket.Socket(Socket.AF_INET, Socket.SOCK)
    Server=('127.0.0.1', 12000)
```

```
while true:
    msg = input("Enter message (or 'quit' to stop): ")
    if msg.lower() == 'quit':
    print("[CLIENT] closing connection...") break
    data, _ = c.recvfrom(1024)
    print("Echo from Server:", data.decode())
c.close()
    print("[CLIENT] Stopped.")
# ... MAIN ...
if __name__ == "__main__":
    threading.Thread(target=udp_server).
    start() udp_client()
```

output:

~~Enter message~~
[Server] Received from('127.0.0.1', 49670): how are you.
Echo from Server: how are you
Enter message (or 'Quit' to Stop): hello.
[Server] Received from ('127.0.0.1', 49670): how are you.
Enter message (or 'quit' to Stop): quit
[Client] closing connection.. [Server] Client('127.0.0.1'
4 9670) ended the connection.

[Client] Stopped. [Server] Stopped.

Result: Implementation of TCP/UDP Successfully executed